

INDUSTRY PROBLEM SOLVING TASK

Big Data – Questions and Answers

CO5-AT1

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1. Retail Data Optimization Problem

Question:

A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Answer:

A suitable big data architecture can be designed as follows:

1. Data Sources: Collect sales data from POS systems and the online platform, inventory data from stores and warehouses, and customer behavior data from websites, mobile applications, and loyalty systems.

2. Data Ingestion: Use Apache Kafka for continuous real-time collection of sales, inventory, and customer events. Batch data can be loaded using ETL tools.
3. Data Storage: Store raw and historical data in a cloud data lake such as Amazon S3 or Azure Data Lake. A data warehouse can be used for structured business reporting.
4. Stream Processing: Use Apache Spark Streaming or Apache Flink to process inventory and sales data in real time.
5. Data Processing: Apache Spark can clean, transform, join, and aggregate large datasets.
6. Analytics and Machine Learning: Use ARIMA, Prophet, or LSTM for demand forecasting. Collaborative filtering or other recommendation models can provide personalized product recommendations.
7. Visualization: Use Power BI or Tableau to display inventory levels, sales trends, and demand forecasts.
8. Output: The system can generate low-stock alerts, forecast future demand, and provide personalized offers to customers.

Data flow:

POS/Website/App → Kafka → Spark/Flink → Data Lake/Data Warehouse → ML Models → Dashboards, Alerts and Recommendations.

2. Social Media Fraud Detection Problem

Question:

A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Answer:

A scalable solution can be designed as follows:

1. Data Sources: Collect posts, comments, likes, shares, login activity, account information, device information, and user interaction data.
2. Real-Time Ingestion: Apache Kafka can receive and distribute billions of events continuously.
3. Stream Processing: Apache Flink or Spark Structured Streaming can process the incoming events in real time or near real time.
4. Storage: Store historical data in HDFS or cloud object storage. NoSQL databases can be used for high-volume user and interaction data.
5. Feature Engineering: Create features such as account age, posting frequency, device patterns, unusual activity, sharing behavior, and network relationships.

6. Machine Learning: Random Forest, Gradient Boosting, or neural networks can classify suspicious accounts. NLP models can analyze posts for misinformation indicators. Graph-based methods can identify suspicious account relationships.
7. Real-Time Scoring: Assign each account or post a risk score based on the model output.
8. Action: High-risk accounts or posts can be flagged for review, restricted, or sent for further verification.
9. Monitoring: Dashboards can show fraud trends and misinformation alerts.

Data flow:

Social Media Events → Kafka → Flink/Spark Streaming → Feature Extraction → ML Models → Risk Score → Alert/Action → Historical Storage.

3. Government Data Platform Problem

Question:

A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Answer:

The following governance and data management strategy can be used:

1. Central Data Lake: Store census, economic, and geospatial datasets in a centralized data lake.
2. Data Integration: Use ETL/ELT pipelines and common formats and APIs to integrate information from different departments.
3. Metadata Management: Maintain a data catalog containing dataset descriptions, owners, formats, quality information, and usage rules.
4. Interoperability: Define common schemas, identifiers, APIs, and standards so different agencies can exchange data consistently.
5. Data Quality: Apply validation, deduplication, completeness, and consistency checks.
6. Governance: Assign data owners and data stewards and define access, retention, classification, and sharing policies.

7. Security: Use encryption, role-based access control, multi-factor authentication, audit logs, and secure networks.

8. Privacy: Protect personally identifiable information through masking, anonymization/pseudonymization, minimum-access principles, and controlled sharing.

9. Monitoring and Compliance: Monitor data access, usage, security events, and compliance continuously.

Result:

The platform provides integrated and reusable government data while maintaining interoperability, privacy, security, quality, and accountability.

4. Banking Fraud Detection Problem

Question:

A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Answer:

A suitable real-time fraud detection pipeline is:

1. Data Sources: Collect transaction records, customer profiles, account history, device information, location, merchant information, and previous fraud records.
2. Streaming Ingestion: Use Apache Kafka to receive transactions continuously with high throughput.
3. Stream Processing: Use Apache Flink or Spark Structured Streaming for low-latency transaction processing.
4. Historical Storage: Store historical transactions in a distributed data lake and use databases or a data warehouse for analysis.
5. Feature Engineering: Calculate transaction amount, transaction frequency, location changes, unusual time patterns, merchant information, and customer spending behavior.

6. Machine Learning: Logistic Regression, Random Forest, XGBoost, or neural networks can classify transactions as legitimate or fraudulent. Anomaly detection can identify new or unusual fraud patterns.

7. Real-Time Scoring: Each transaction receives a fraud probability or risk score.

8. Decision Layer: High-risk transactions can be blocked or sent for verification, while low-risk transactions can be approved.

9. Model Improvement: Confirmed fraud cases are stored and used to retrain the models.

Data flow:

Transactions → Kafka → Flink/Spark → Feature Engineering → ML Model → Fraud Score → Approve/Block/Review → Data Lake → Model Retraining.

5. Agriculture Smart Farming Problem

Question:

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Answer:

A precision agriculture solution can be designed as follows:

1. Data Sources: Collect soil moisture, temperature, humidity, rainfall, crop health, drone images, satellite data, irrigation information, and weather data.
2. IoT Data Collection: Sensors installed across fields continuously generate data. Edge devices can collect and preprocess data near the farm.
3. Data Ingestion: Use IoT gateways and MQTT/Kafka to transfer sensor and drone data to the cloud.
4. Storage: Store sensor, image, and historical data in a cloud data lake.
5. Stream Processing: Spark Streaming or Flink can process real-time sensor readings and generate alerts.
6. Analytics: Descriptive analytics shows current field conditions. Predictive analytics can forecast crop yield, irrigation requirements, pest risks, and weather effects.

7. Machine Learning: Regression models can predict yield and water requirements. Classification or deep learning models can detect crop diseases from drone images.

8. Decision Support: The system can recommend when to irrigate, how much water or fertilizer to use, and which areas need attention.

9. Visualization: Dashboards and mobile applications can display field maps, alerts, crop health, and recommendations.

Data flow:

Sensors/Drones/Weather → IoT Gateway → Kafka → Cloud Data Lake → Spark/Flink → ML & Analytics → Alerts/Recommendations → Farmer Dashboard.